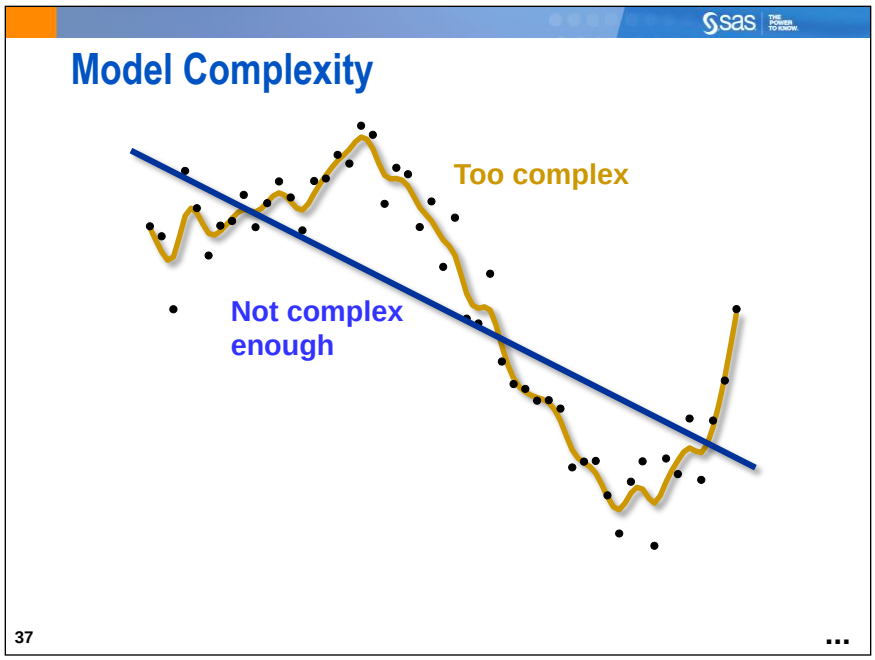
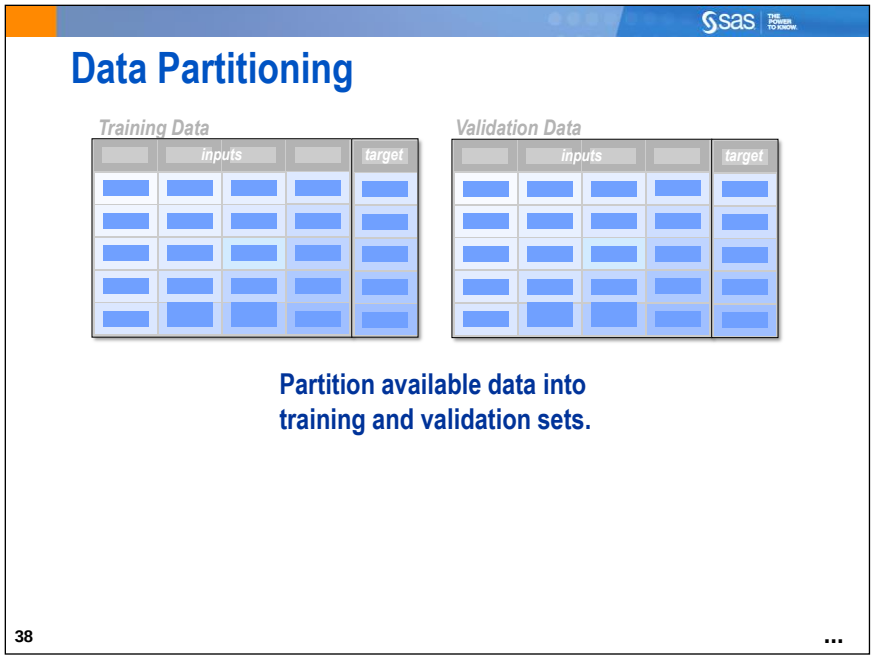


Fitting a model to data requires searching through the space of possible models. Constructing a model with good generalization requires choosing the right complexity.

Selecting model complexity involves a trade-off between bias and variance. An insufficiently complex model might not be flexible enough, which can lead to *underfitting*, that is, systematically missing the signal (high bias).

A naïve modeler might assume that the most complex model should always outperform the others, but this is not the case. An overly complex model might be too flexible, which can lead to *overfitting*, that is, accommodating nuances of the random noise in the particular sample (high variance). A model with the right amount of flexibility gives the best generalization.



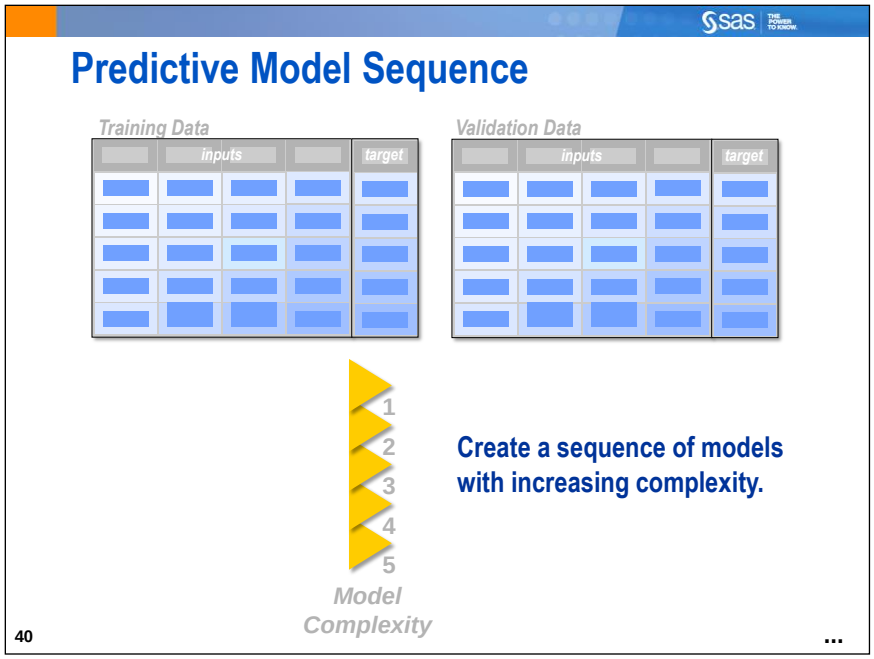
A quote from the popular introductory data analysis textbook *Data Analysis and Regression* by Mosteller and Tukey (1977) captures the difficulty of properly estimating model complexity.

> "Testing the procedure on the data that gave it birth is almost certain to overestimate performance, for the optimizing process that chose it from among many possible procedures will have made the greatest use of any and all idiosyncrasies of those particular data…."

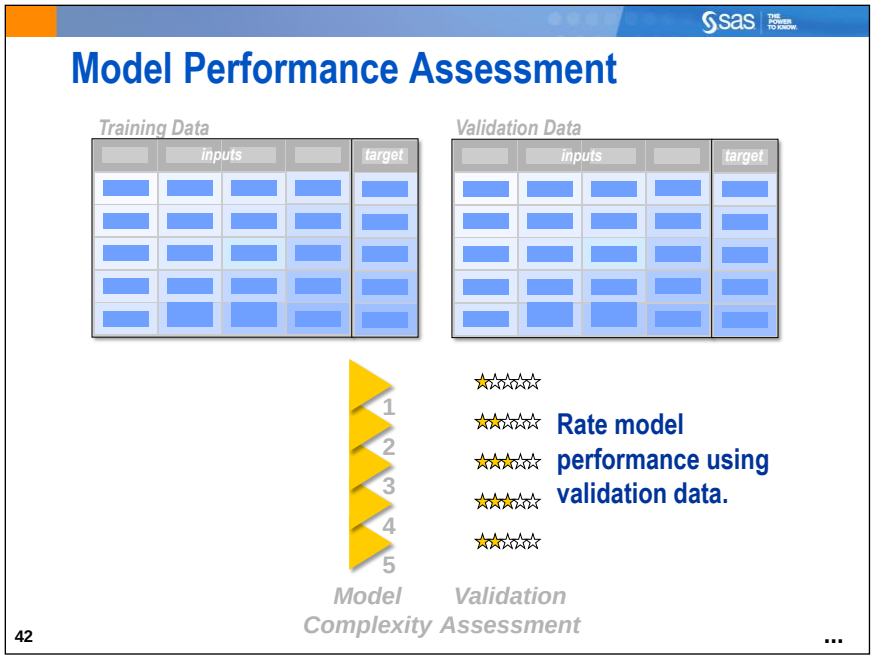
In predictive modeling, the standard strategy for honest assessment of model performance is *data splitting*. A portion is used for fitting the model, that is, the training data set. The remaining data is separated for empirical validation.

The *validation data set* is used for monitoring and tuning the model to improve its generalization. The tuning process usually involves selecting among models of different types and complexities. The tuning process optimizes the selected model on the validation data.

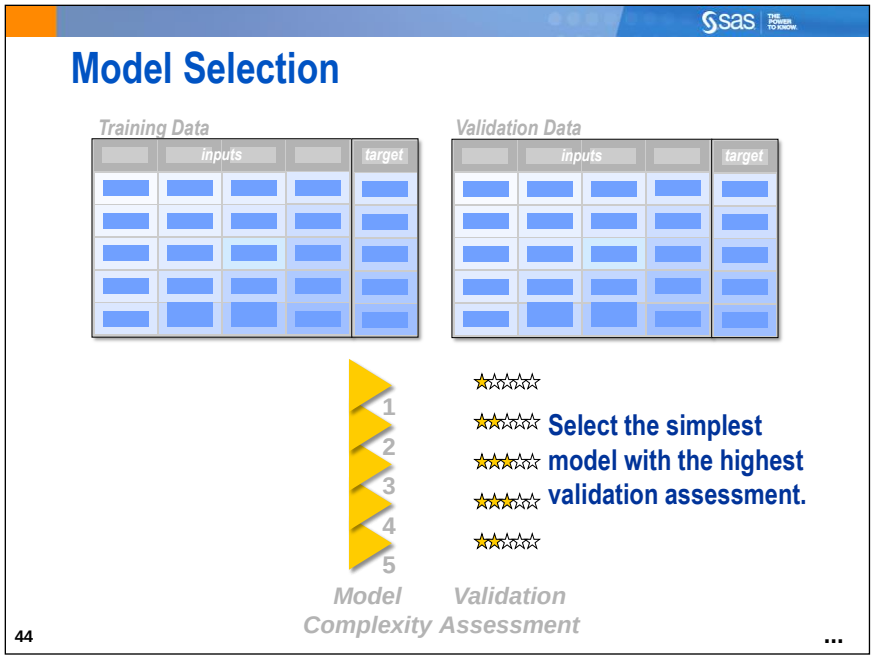
With small or moderate data sets, data splitting is inefficient; the reduced sample size can severely degrade the fit of the model. Computer-intensive methods, such as the *cross validation* and *bootstrap* methods, were developed so that all the data can be used for both fitting and honest assessment. However, data mining usually has the luxury of massive data sets.



In the modern predictive approach, the strategy for choosing model complexity is to build a sequence of similar models using the training component only. The models are usually ordered with increasing complexity.



As Mosteller and Tukey caution in the earlier quotation, using performance on the training data set usually leads to selecting a model that is too complex. (The classic example is selecting linear regression models based on R square.) To avoid this problem, the best practice is to select the model from the sequence based on validation data performance.



In keeping with Ockham's razor, the best model is the simplest model with the highest validation performance.